FIITJEE

ALL INDIA TEST SERIES

FULL TEST – II

JEE (Main)-2024

TEST DATE: 07-01-2024

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

PART – A

SECTION – A
(One Options Correct Type)

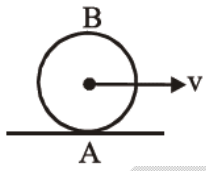
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A bus is moving with a velocity of 10 m/s on a straight road. A scooterist wishes to overtake the bus in 100 seconds. If the bus is at a distance of 1 km form the scooterist, at what velocity should the scooterist chase the bus:
(A) 50 m/sec (B) 40 m/sec
(C) 30 m/sec (D) 20 m/sec

2. The length of an elastic string is x when the tension is 5N. Its length is y when the tension is 7N. What will be its length, when the tension is 9N?
(A) 2y + x (B) 2y – x
(C) 7x – 5y (D) 7x + 5y

3. A rod of length L is placed on x-axis between x = 0 and x = L. The linear density i.e., mass per unit length denoted by ρ, of this rod, varies as, $\rho = a + bx$. What should be the dimensions of b?
(A) $M^2L^1T^0$ (B) $M^1L^{-2}T^0$
(C) $M^{-1}L^3T^1$ (D) $M^{-1}L^2T^3$

4. A wheel is rolling on a plane road. The linear velocity of centre of mass is v. Then velocities of the points A and B on circumference of wheel relative to road will be:



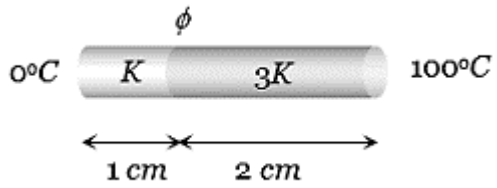
(A) $v_A = v,\ v_B = 0$ (B) $v_A = v_B = 0$
(C) $v_A = 0,\ v_B = v$ (D) $v_A = 0,\ v_B = 2v$

5. A metallic wire of density d is lying horizontal on the surface of water (surface tension = t). The maximum length of wire so that it may not sink will be:
(A) $\sqrt{\frac{2Tg}{\pi d}}$ (B) $\sqrt{\frac{2\pi T}{dg}}$
(C) $\sqrt{\frac{2T}{\pi dg}}$ (D) any length

6. Two points of a rod move with velocities 3v and v perpendicular to the rod and in the same direction, separated by a distance r. Then the angular velocity of the rod is:
(A) 3v/r (B) 4v/r
(C) 5v/r (D) 2v/r

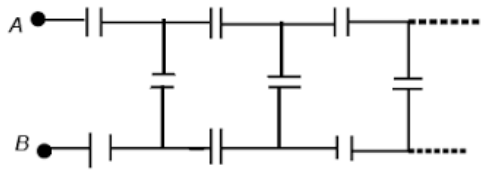
7. For hydrogen gas $C_p - C_v = a$ and for oxygen gas $C_p - C_v = b$, so the relation between a and b is given by:
(A) a = 16b (B) 16a = b
(C) a = 4b (D) a = b

8. Two bars of thermal conductivities K and 3K and lengths 1 cm and 2 cm, respectively, have equal cross-sectional area. They are joined length wise as shown in the figure. If the temperature at the ends of this composite bar is 0°C and 100°C, respectively (see figure), then the temperature ϕ of the interface is:



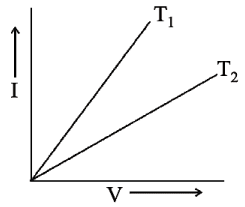
(A) 50°C
(B) 100/3°C
(C) 60°C
(D) 200/3°C

9. The equivalent capacitance of the network, (with all capacitors having the same capacitance C)



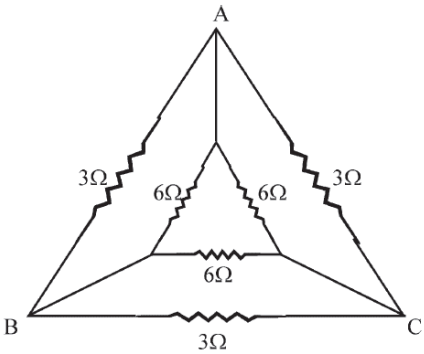
(A) ∞
(B) Zero
(C) $C\left[\left(\sqrt{3}-1\right)/2\right]$
(D) $C\left[\left(\sqrt{3}+1\right)/2\right]$

10. The current I vs voltage V graphs for a given metallic wire at two difference temperatures T_1 and T_2 are shown in the figure. It is concluded that:



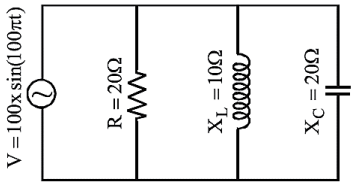
(A) $T_1 > T_2$
(B) $T_1 < T_2$
(C) $T_1 = T_2$
(D) $T_1 = 2T_2$

11. In the circuit shown the effective resistance between B and C is:



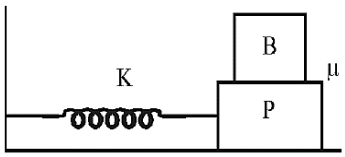
(A) 3 Ω
(B) 4 Ω
(C) 4/3 Ω
(D) 3/4 Ω

12. In the given circuit, the current drawn from the source is:



(A) 20 A
(B) 10 A
(C) 5 A
(D) $5\sqrt{2}$ A

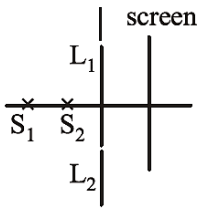
13. A flat plate P of mass 'M' executes SHM in a horizontal plane by sliding over a frictionless surface with a frequency υ. A block 'B' of mass 'm' rests on the plate as shown in figure. Coefficient of friction between the surface of B and P is μ. What is the maximum amplitude of oscillation that the plate block system can have if the block B is not to slip on the plate:



(A) $\frac{\mu g}{4\pi^2\upsilon^2}$
(B) $\frac{\mu g}{\pi^2\upsilon^2}$
(C) $\frac{\mu g}{8\pi^2\upsilon^2}$
(D) $\frac{\mu g}{2\pi^2\upsilon^2}$

14. A glass slab has the left half of refractive index n_1, and the right half of $n_2 = 3n_1$. The effective refractive index of the whole slab is:
(A) $n_1/2$
(B) $2n_1$
(C) $3n_1/2$
(D) $2n_1/3$

15. In the arrangement shown L_1, L_2 are slits and S_1, S_2 two independent sources on the screen, interference fringes:



(A) Will not be there
(B) Will not be there if the intensity of light reaching the screen from S_1 and S_2 are equal.
(C) Will be there under all circumstances.
(D) We will have only the central fringe.

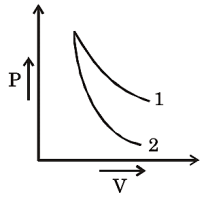
16. What is the ratio of the circumference of the first Bohr orbit for the electron in the hydrogen atom to the de Broglie wavelength of electrons having the same velocity as the electron in the first Bohr orbit of the hydrogen atom?
(A) 1:1
(B) 1:2
(C) 1:4
(D) 2:1

17. An electric field is expressed as $\vec{E} = 2\hat{i} + 3\hat{j}$. Find the potential difference (V_A - V_B) between two points A and B whose position vectors are given by $r_A = \hat{i} + 2\hat{j}$ and $r_B = 2\hat{i} + \hat{j} + 3\hat{k}$.

(A) $-1V$ (B) $1V$

(C) $2V$ (D) $3V$

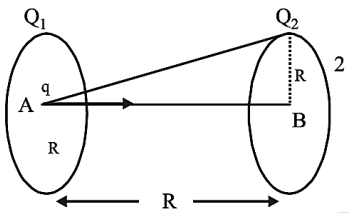
18. P – V plots for two gases during adiabatic processes are shown in the figure. Plots 1 and 2 should correspond respectively to:



(A) He and Ar (B) He and O_2

(C) O_2 and N_2 (D) O_2 and He

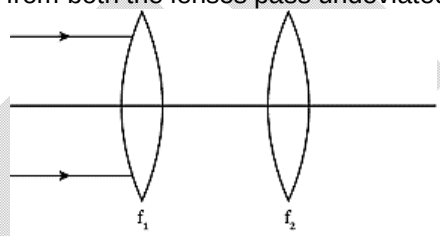
19. Two identical thin rings, each of radius R metres, are coaxially placed at a distance R metres apart. If Q_1 coulomb and Q_2 coulomb are respectively, the charges uniformly spread on the two rings, the work done in moving a charge q from the centre of one ring to that of the other is:



(A) Zero (B) $q(Q_1 - Q_2)\left(\sqrt{2} - 1\right) / \sqrt{2}\ 4\pi\varepsilon_0 R$

(C) $q\sqrt{2}(Q_1 + Q_2) / 4\pi\varepsilon_0 R$ (D) $q(Q_1 + Q_2)\left(\sqrt{2} + 1\right) / \sqrt{2} 4\pi\varepsilon_0 R$

20. A parallel beam of light is incident on the system of two convex lenses of focal lengths f_1 = 20 cm and f_2 = 10 cm. What should be the distance between the two lenses so that rays after refraction from both the lenses pass undeviated?



(A) 60 cm (B) 30 cm

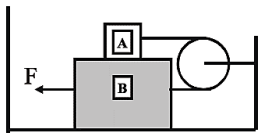
(C) 90 cm (D) 40 cm

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

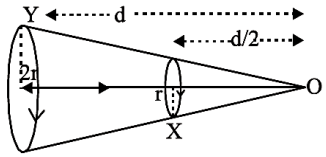
21. The masses of the blocks A and B are 0.5 kg and 1 kg respectively. These are arranged as shown in the figure and are connected by a massless string. The coefficient of friction between all contact surfaces is 0.4. The force (in N) necessary to move the block B with constant velocity will be: ($g = 10\ m/s^2$):



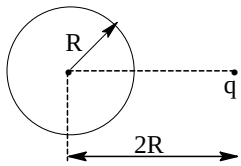
22. An artificial satellite is moving in a circular orbit around the earth with a speed equal to half of the magnitude of escape velocity from the earth. If the satellite is stopped suddenly in its orbit and allowed to fall freely onto the earth, then speed (in ms^{-1}) with which it hits the surface of the earth is $\sqrt{kgR}$. Find the value of k. (Given, g = 9.8 ms^{-2} and R = 6.4 × 10^6 m)

23. A non-conducting partition divides a container into two equal compartments. One is filled with helium gas at 200K and the other is filled with oxygen gas at 400K. The number of molecules in each gas is the same. If the partition is removed to allow the gases to mix, the final temperature (in K) will be:

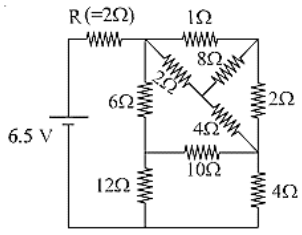
24. Two circular coils X and Y, having equal number of turns, carry equal currents in the same sense and subtend same sense angles at point O. If the smaller coil X is midway between O and Y, then if we represent the magnetic induction due to bigger coil Y at O as B_Y and due to smaller coil X at O as B_X then the ratio $\frac{B_X}{B_Y}$ is:



25. A point charge is placed outside a spherical shell as shown in the figure. Find the electric potential (in kilo volt) on the surface of shell (Take: $q=1\mu c \text{ and } R=50cm$)



26. In the following circuit, the current through the resistor $R(=2\Omega)$ is I amperes. The value of I is_____



27. The pressure and volume of an ideal gas are related as $P=\frac{a}{1+\left(\frac{V}{b}\right)^3}$, where 'a' and 'b' are constants. The translational kinetic energy due to the thermal motion of the gas sample at volume $V=b$, (given that $ab=12\,kJ$) is $n\,kJ$. Find n.

28. The potential energy of a particle is determined by the expression $U=\alpha\left(x^2+y^2\right)$, where α is a positive constant. The particle begins to move from a point with the co-ordinates $(3,3)$ only under the action of potential fields force. When it reaches the point $(1,1)$ its kinetic energy is $4K\alpha$. Find the value of K.

29. A plane progressive wave is given by x = (40 cm) cos (50πt – 0.02πy) where y is in cm and t in s. the particle velocity at y = 25 cm in time t = 1/100s will be $10\pi\sqrt{n}$ m/s. What is the value of n.

30. A charge of 5 μC is placed at the centre of a square ABCD of side 10 cm. Find the work done (in μJ) in moving a charge of 1 μC from A to B.

Chemistry

PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. An aqueous solution of aniline of concentration 0.24M is prepared. What is concentration of sodium hydroxide is needed in this solution so that anilinium ion concentration remains at 1 x 10^{-8} M? (K_b for $C_6H_5NH_3^+ = 2.4 \times 10^{-5}$M)

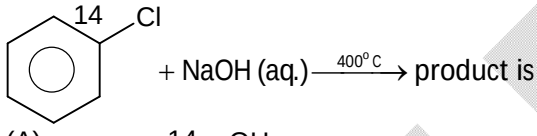
(A) 10^{-2}M (B) 10^{-1}M

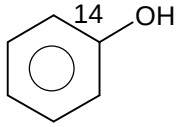
(C) 10^{-3}M (D) 10^{-4}M

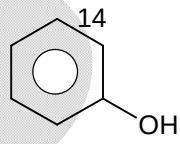
32. For a cell reaction involving two electron charge, the standard emf of the cell is 0.295 V at 25°C. The equilibrium constant of the reaction at 25°C will be:

(A) 1 x 10^{10} (B) 10

(C) 29.5 x 10^{-2} (D) 2.95 x 10^{-10}

33. 

(A) 

(B) 

(C) Both (A) and (B) (D) No reaction

34. The standard heat of formation listed for gaseous NH_3 is – 11.02 kcal/mol at 298 K. Given that at 298 K, the constant pressure heat capacities of gaseous N_2, H_2 and NH_3 are respectively 6.96, 6.89, 8.38 cal/mol. ΔH_{773K} for the reactions,

$$\frac{1}{2}N_2(g)+\frac{3}{2}H_2(g)\rightarrow NH_3(g)$$

(A) $-10.6\,kcal\,mol^{-1}$ (B) $-13.6\,kcal\,mol^{-1}$

(C) $-12.4\,kcal\,mol^{-1}$ (D) $-11.8\,kcal\,mol^{-1}$

35. A cobalt amine has the formula $Co(NH_3)_6Cl_3$, with $AgNO_3$ solution, one-third of the chloride is precipitated. The complex shows cis-trans isomerism. It can have the structure:

(A) $[Co(NH_3)_6]Cl_3$ (B) $[Co(NH_3)_5Cl]Cl_2$

(C) $[Co(NH_3)_4Cl_2]Cl$ (D) $[Co(NH_3)_5.H_2O]Cl_3$

36. The pair in which both species have same magnetic moment is:

(A) $\left[Cr(H_2O)_6\right]^{2+}, \left[CoCl_4\right]^{2-}$ (B) $\left[Cr(H_2O)_6\right]^{2+}, \left[Fe(H_2O)_6\right]^{2+}$

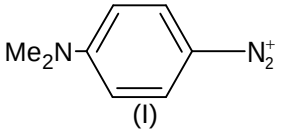
(C) $\left[Mn(H_2O)_6\right]^{2+}, \left[Cr(H_2O)_6\right]^{2+}$ (D) $\left[CoCl_4\right]^{2-}, \left[Fe(H_2O)_6\right]^{2+}$

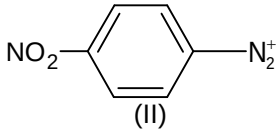
37. Identify the product "Z" in the following series of reactions:

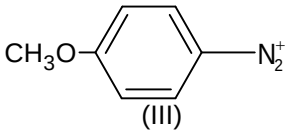
$C_6H_{12}O_6$ (glucose) $\xrightarrow{HCN} X \xrightarrow{H_2O} Y \xrightarrow{HI} Z$:

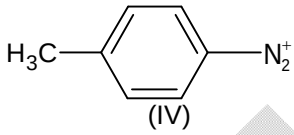
(A) hexanoic acid
(B) α-methyl caproic acid
(C) Heptanoic acid
(D) none of these

38. Consider the following diazonium ions:


(I)


(II)


(III)


(IV)

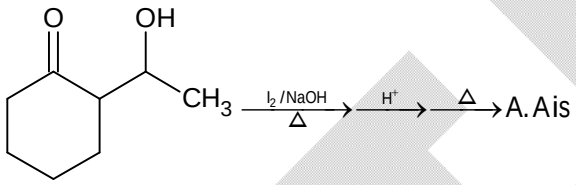
The order of reactivity toward diazo-coupling with phenol in the presence of dil. NaOH is

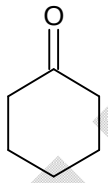
(A) I < IV < II < III
(B) I < III < IV < II
(C) III < I < II < IV
(D) III < I < IV < III

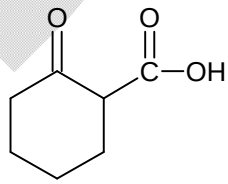
39. which one is the best method of reducing 3-bromoproponal to 1-bromopropane?

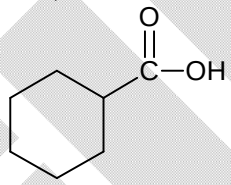
$BrCH_2CH_2CHO \rightarrow BrCH_2CH_2CH_3$

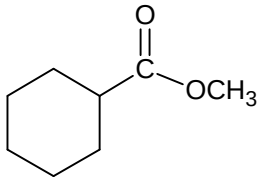
(A) Wolf-Kishner reduction
(B) Clemmensen reduction
(C) Either of the two
(D) None of the two

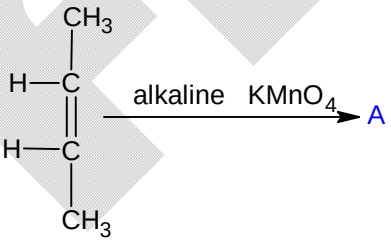
40. 

(A) 

(B) 

(C) 

(D) 

41. 

Which is true about this reaction?

(A) A is meso 1, 2-butan-di-ol formed by syn addition
(B) A is meso 1, 2-butan-di-ol formed by anti addition
(C) A is racemic mixture of d and ℓ, 1, 2-butan-di-ol formed by anti addition
(D) A is racemic mixture of d and ℓ, 1, 2-butan-di-ol formed by syn addition

42. The incorrect statement regarding $O(SiH_3)_2$ and OCl_2 molecule is/are:
(A) The strength of back bonding is more in $O(SiH_3)_2$ molecule than OCl_2 molecule
(B) $Si-\hat{O}-Si$ bond angle in $O(SiH_3)_2$ is greater than $Cl-\hat{O}-Cl$ bond angle in OCl_2
(C) Then nature of back bond in both molecules is $(2p_\pi - 3d_\pi)$
(D) Hybridisation of central O-atom in both molecules is same

43. Ammonia forms the complex $[CuCNH_3)_4]^{2+}$ with copper ions in alkaline solution but not in acidic solution. The reason for this is:
(A) In alkaline solution $Cu(OH)_2$ is precipitated which is soluble in excess of alkali
(B) Copper hydroxide is amphoteric substance
(C) In acidic solution hydration protects Cu^{2+} ions
(D) In acidic solution protons are coordinated with ammonia molecules forming NH_4^+ ions

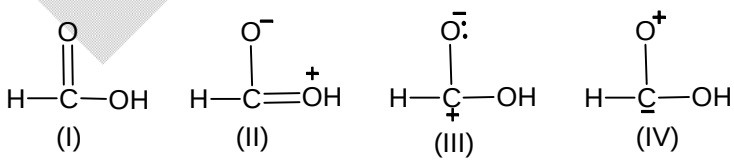
44. The hydrolysis constant for $ZnCl_2$ will be:
(A) $K_h = \frac{K_W}{K_b}$ (B) $K_h = \frac{K_W^2}{K_b}$
(C) $K_h = \frac{K_W^2}{K^2{}_b}$ (D) $K_h = \frac{K_b}{K^2{}_W}$
Where K_b is effective dissociation constant of base Zn^{++}

45. Consider the following conversions:
(i) $O_{(g)} + e^- \rightarrow O^-_{(g)},\ \Delta H_1$ (ii) $F_{(g)} + e^- \rightarrow F^-_{(g)},\ \Delta H_2$
(iii) $Cl_{(g)} + e^- \rightarrow Cl^-_{(g)},\ \Delta H_3$ (iv) $O^-_{(g)} + e^- \rightarrow O^{2-}_{(g)},\ \Delta H_4$
That according to given information the incorrect statements is:
(A) ΔH_3 is more negative than ΔH_1 and ΔH_2
(B) ΔH_1 is less negative than ΔH_2
(C) ΔH_1, ΔH_2 and ΔH_3 are negative whereas ΔH_4 is positive
(D) ΔH_1 and ΔH_3 are negative whereas ΔH_2 and ΔH_4 are positive

46. What is the geometry of the IBr_2^- ion?
(A) Linear (B) Bent shape with bond angle of about 90°
(C) Bent shape with bond angle of about 109° (D) Bent shape with bond angle of about 120°

47. The formation of PH_4^+ is difficult compared to NH_4^+ because:
(A) lone pair of phosphorus is optically inert
(B) lone pair of phosphorus resides in almost pure p-orbital
(C) lone pair of phosphorus resides at sp^3 orbital
(D) lone pair of phosphorus resides in almost pure s-orbital

48. Which of the following solid salt on heating with solid $K_2Cr_2O_7$ and cone. H_2SO_4 orange red vapours are evolved which tum aqueous NaOH solution yellow?
(A) NaBr (B) NaCl
(C) $NaNO_3$ (D) NaI

49. 
(I) (II) (III) (IV)
Increasing order of stability is
(A) I < III < II < IV (B) IV < III < II < I
(C) III < IV < II < I (D) II < IV < III < I

50. For the electrochemical cell, $M|M^+||X^-|X, E°M^+/M = 0.44V$ *and* $E°\left(X/X^-\right) = 0.33V$. From this data one can deduce that:

(A) $M + X \rightarrow M^+ + X^-$ is the spontaneous reaction.

(B) $M^+ + X^- \rightarrow M + X$ is the spontaneous reaction.

(C) $E_{cell} = 0.77V$

(D) $E_{cell} = -0.77V$

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. At 273K and atm, 'a' litre of N_2O_4 decompose to NO_2 as: $N_2O_4 \rightleftharpoons 2NO_2(g)$. To what extent has the decomposition (%) proceeded when the original volume is 25% less than that of existing volume?

52. For a first order reaction, calculate the ratio between the time taken to complete three-fourth of the reaction between the time taken to complete half of the reaction.

53. Suppose 10^{-17} J of light energy is needed by the interior of the human eye to see an object. How many photons of green light $(\lambda = 550\text{nm})$ are needed to generate this minimum amount of energy?

54. $t_{1/2}$ of a reaction, $A \rightarrow \text{Product}\left(\text{order} = \frac{3}{2}\right)\text{is represented by}\, t_{1/2} \propto \frac{1}{(a_0)^m}$. The value of 2m is:

55. The energy of activation of reaction, if its rate double when the temperature is raised from 290 K to 300 K is x kcal then the value of x/3 is _____ (Take R = 2 cal/k/mol)

56. In a constant volume calorimeter, 3.5 g of gas with molar mass 28 g mol^{-1} was burn in excess oxygen at 298.0 K. The temperature of the calorimeter was found to increase from 298.0 K to 298.45 K due to the combustion process. Given that the heat capacity of the calorimeter is $2.5\,kJK^{-1}$, the numerical value for the enthalpy of combustion of the gas in $kJ\,mol^{-1}$ is:

57. The pH of an aqueous solution of 0.1 M solution of a weak monoprotic acid which is 1% ionized is:

58. Find out the % of oxalate ion in given sample of oxalate salt of which 0.3 g is present in 100 mL of solution required 90 mL of N/20 $KMnO_4$ for complete oxidation.

59. The molal freezing point constant of C_6H_6 is 4.90 and its melting point is 5.51°C. A solution of 0.816 g of compound A when dissolved in 7.5 g of benzene freezes at 1.59°C. Calculate molar mass of compound A.

60. Find the volume of a 0.2 M solution of MnO_4^- that will react with 50 mL of 0.1 M solution of $C_2O_4^{2-}$ in acidic medium____

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. The number of integer values of a for which the inequality $x^2 + ax + a^2 + 6a < 0$ is satisfied for all $x \in (1, 2)$ is
(A) 5 (B) 7
(C) 8 (D) 6

62. The relation R defined on the set A = {1, 2, 3, 4, 5} is given by R = (x : y): $|x^2 - y^2| < 16$ then
(A) R is reflexive and symmetric (B) R is transitive
(C) R is not symmetric (D) R is reflexive and transitive

63. A number is said to be a nice number if it has exactly 4 factors. (Including one and number itself). Let $n = 2^3 \times 3^2 \times 5^3 \times 7 \times 11^2$, then number of factors, which are nice numbers is
(A) 36 (B) 12
(C) 10 (D) 9

64. The value of $\lim\limits_{x\to\infty} \frac{x^n + nx^{n-1} + 1}{e^{[x]}}, n \in I$ is (where [.] denotes greatest integer function)
(A) 1 (B) zero
(C) n (D) n(n –1)

65. The solution of the differential equation $(1 - xy - x^5y^5)\,dx - x^2(x^4y^4 + 1)dy = 0$ is given by
(A) $x = ce^{xy + \frac{1}{5}x^5y^5}$ (B) $x = ce^{xy - \frac{1}{5}x^5y^5}$
(C) $x = ce^{x^2y^2 + \frac{1}{5}x^5y^5}$ (D) $x = ce^{x^2y^2 - \frac{1}{5}x^5y^5}$

66. The number of point(s) on the line $3x + 4y = 5$, which are at a distance of $\sec^2\theta + 2\,\text{cosec}^2\theta$, $\theta \in R$, from the point (1, 3) is
(A) 1 (B) 2
(C) 3 (D) zero

67. The equation of circumcircle of an equilateral triangle is $x^2 + y^2 + 2gx + 2fy + c = 0$ and one vertex of the triangle is (1, 1). The equation of incircle of the triangle is:
(A) $4(x^2 + y^2) = g^2 + f^2$
(B) $4(x^2 + y^2) + 8gx + 2fy + c = (1 - g)(1 + 3g) + (1 - f)(1 + 3f)$
(C) $4(x^2 + y^2) + 8gx + 8fy = g^2 + f^2$
(D) $4(x^2 + y^2) + 8(g^2 + f^2) = 0$

68. If AFB is a focal chord of the parabola $y^2 = 4ax$ and AF = 4, FB = 5 (F being the focus of given parabola) then the latus-rectum of the parabola is equal to
(A) $\frac{80}{9}$ (B) $\frac{9}{80}$
(C) 9 (D) 80

69. If the ellipse $\frac{x^2}{16}+\frac{y^2}{b^2}=1$ and the hyperbola $\frac{x^2}{144}-\frac{y^2}{81}=\frac{1}{25}$ are orthogonal then the value of b^2

(A) 1 (B) 5
(C) 7 (D) 9

70. If $\cos^2\theta = \frac{1}{3}\left(a^2-1\right)$ and $\tan^2\frac{\theta}{2} = \tan^{2/3}\alpha$, then $\cos^{2/3}\alpha + \sin^{2/3}\alpha$ =

(A) $2a^{2/3}$ (B) $\left(\frac{2}{a}\right)^{2/3}$
(C) $\left(\frac{2}{a}\right)^{1/3}$ (D) $2a^{1/3}$

71. A and B play a game of tennis. The situation of the game is as follows; if one scores two consecutive points after a deuce he wins; if loss of a point is followed by win of a point, it is deuce. The chance of a server to win a point is 2/3. The game is at deuce and A is serving. Probability that A will win the match is, (serves are changed after each point scored)

(A) 3/5 (B) 2/5
(C) 1/2 (D) 4/5

72. If A, B, C are the sets of all positive divisors of 10^{30}, 20^{20} and 30^{10} respectively. Then $(A \cup B \cup C)$ equals

(A) 2381 (B) 13981
(C) 4381 (D) 7161

73. If in a moderately asymmetrical distribution, mode and mean of the data are 6λ and 9λ respectively, then median is

(A) 8λ (B) 7λ
(C) 6λ (D) 5λ

74. 'n_1' men and 'n_2' women are to be seated in a row so that no two women sit together. If $n_1 > n_2$ then total number of ways in which they can be seated, is equal to:

(A) ${}^{n_1}C_{n_2}$ (B) ${}^{n_1}C_{n_2}\ (n_1!)(n_2!)$
(C) ${}^{n_1}C_{n_2+1}\ (n_1!)(n_2!)$ (D) ${}^{n_1+1}C_{n_2}\ (n_1!)(n_2!)$

75. If coefficient of $x^2y^3z^4$ in $(x+y+z)^n$ is A, (where $A \neq 0$) then coefficient of x^4y^4z is

(A) 2A (B) $\frac{nA}{2}$
(C) $\frac{A}{2}$ (D) nA

76. Consider the equation $\frac{xy}{x+y}=2^3 3^4 5^6$ then the number of positive integral solutions of the equation are

(A) 140 (B) 819
(C) 72 (D) 601

77. $I=\int \frac{(1+x)}{x(1+xe^x)^2}\,dx$ is equal to

(A) $\ln\left(\frac{xe^x}{1+xe^x}\right)-\left(\frac{1}{1+xe^x}\right)+c$

(B) $\ln\left(\frac{xe^x}{1+xe^x}\right)+\left(\frac{1}{1-xe^x}\right)+c$

(C) $\ln\left(\frac{xe^x}{1-xe^x}\right)+\left(\frac{1}{1+xe^x}\right)+c$

(D) $\ln\left(\frac{xe^x}{1+xe^x}\right)+\left(\frac{1}{1+xe^x}\right)+c$

78. $\lim\limits_{n\to\infty}\left[\tan\frac{\pi}{3n}+\tan\frac{2\pi}{3n}+..+\tan\frac{\pi}{3}\right]\frac{1}{n}=$

(A) $\frac{3}{\pi}\ln 2$

(B) $\frac{2}{\pi}\ln 2$

(C) $\frac{3}{\pi}\ln 3$

(D) $\frac{3}{\pi}$

79. Let $\vec{a},\vec{b},\vec{c}$ be three non zero vectors such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$ Then $\lambda\vec{b}x\vec{a}+\vec{b}x\vec{c}+\vec{c}x\vec{a}=\vec{0}$, where λ is equal to

(A) 1

(B) 2

(C) –1

(D) –2

80. If $\vec{a}=x\hat{i}+(x-1)\hat{j}+\hat{k}$ and $\vec{b}=(x+1)\hat{i}+j+a\hat{k}$ always make an acute angle with each other for every value of $x\in R$, then

(A) $a\in(-\infty,2)$

(B) $a\in(2,\infty)$

(C) $a\in(-\infty,1)$

(D) $a\in(1,\infty)$

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Let ABC be a triangle. A be the point (1, 2), y = x is the perpendicular bisector of AB and $x-2y+1=0$ is the angle bisector of angle C. If the equation of BC is given by $ax+by-5=0$, then the value of a + b is:

82. If a, b, c are in H.P., then the value of $\frac{b+a}{b-a}+\frac{b+c}{b-c}$ is equal to

83. The tangents drawn from the origin to the circle $x^2+y^2-2rx-2hy+h^2=0$ are perpendicular then the value of $|r^2-h^2|$ is ___

84. If Minimum distance between the curve $y^2=4x$ and $x^2+y^2-12x+31=0$ is equal to λ, then λ^2 is equal to

85. If The area of the region consisting of points (x, y) satisfying $|x\pm y|\le 2$ and $x^2+y^2\ge 2$ is equal to $(k_1-k_2\pi)$ sq. unit, then value of (k_1-k_2) is where $k_1, k_2\in N$

86. A square matrix A_n is defined as $A_n = [a_{ij}]_{n \times n'}$, where $a_{ij} = \begin{cases} \frac{1}{i^2+j^2} & \text{if } i \le j \\ 0 & \text{if } i > j \end{cases}$, then the value of $\frac{1}{10}\left(\sum_{n=1}^{5} \text{trace of}\left(A_n^{-1}\right)\right)$ is equal to

87. Four people sit around a circular table, and each person will roll a normal six sided die once. The probability that no two people sitting next to each other will roll the same number is $\frac{N}{1296}$. Then the value of $\left[\frac{N}{10}\right]$ is (where [.] denotes the greatest integer function)

88. In a quadrilateral ABCD, $\overrightarrow{AC}$ is the bisector of the $\left(\overrightarrow{AB}\hat{}\,\overrightarrow{AD}\right)$ which is $\frac{2\pi}{3}$, $15\,|\overrightarrow{AC}| = 3|\overrightarrow{AB}| = 5|\overrightarrow{AD}|$, if $\cos^2\left(\overrightarrow{BA}\hat{}\,\overrightarrow{CD}\right) = \frac{p}{q}$ (where p and q are co-prime and $\vec{a}\hat{}\,\vec{b}$ denotes the angle between $\vec{a}$ and $\vec{b}$) then p + q is equal to.

89. Let $\vec{u}$ be a vector on rectangular coordinate system with sloping 60°. Suppose that $\left|\vec{u}-\hat{i}\right|$ is geometric mean of $\left|\vec{u}\right|$ and $\left|\vec{u}-2\hat{i}\right|$ where $\hat{i}$ is the unit vector along x-axis then $\left|\vec{u}\right|$ has the value equal to $\sqrt{a}-\sqrt{b}$ where a, $b \in N$, then the value $(a+b)^3+(a-b)^3$ is

90. Consider a system of equations $x^4 + y^4 - z^4 = 1$, $x^4 + \lambda y^4 + z^4 = \lambda + 2$ and $x^4 - y^4 + \lambda z^4 = \lambda$, where λ is a real parameter. If the number of real solutions of the system of equations is n, then n is equal to